

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Computer Vision Project Report

**Sparse-View RGB 3D Scene Reconstruction
with Fine-Tuned Estimated-Depth Correction**

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Hanoi, June 25, 2026

Contents

1	Introduction	1
2	Background and Related Work	2
2.1	Sparse-View RGB-Only 3D Reconstruction	2
2.2	Monocular Depth Estimation	3
2.3	Depth-Assisted 3D Reconstruction	3
2.4	Project Positioning	3
3	Model Architectures Used in the Project	3
3.1	MV-DUS _t 3R+ Backbone	4
3.2	Depth Anything V2 Metric Indoor	5
4	Problem Formulation	5
5	Proposed Method	7
5.1	Pipeline Overview	7
5.2	Back-Projection from Estimated Depth	8
5.3	Residual-Gated Correction	9
6	Dataset and Experimental Setting	9
6.1	Full Frame Pool for Depth Fine-Tuning	10
6.2	Controlled Reconstruction Subset	10
6.3	Sparse-View Group Construction	11
6.4	Sparse-View Selection Protocol	11
6.5	Use of Measured Depth	12
6.6	Coverage Terminology and Benchmark Boundary	13
7	Evaluation Metrics	13
7.1	Depth Estimation Metrics	13
7.2	Point-Cloud Reconstruction Metrics	14
8	Results	15
8.1	Depth Estimator Fine-Tuning	15
8.2	3D Reconstruction Results	17
8.3	Ablation Study	18
8.4	Qualitative Analysis	19
9	Discussion	19
10	Conclusion	20

1. Introduction

Three-dimensional scene reconstruction is a fundamental problem in computer vision. It supports applications such as augmented reality, robotics, digital twins, indoor mapping, visual localization, and 3D content creation. Classical reconstruction pipelines usually require many images, calibrated cameras, reliable pose estimation, feature matching, structure-from-motion, and multi-view stereo. Although these systems can produce high-quality geometry, they are difficult to deploy when only a small number of ordinary RGB images are available.

In practical settings, the input is often sparse. A user may only have three to five images of an indoor scene, captured by a standard phone camera. These images usually contain RGB appearance only. They do not provide measured depth maps, camera intrinsics, or camera poses. If a reconstruction system requires depth at inference time, it becomes tied to RGB-D cameras, LiDAR devices, or specialized scanning setups. This requirement is not compatible with most everyday image collections.

Recent feed-forward reconstruction models such as MV-DUST3R+ reduce this dependency by reconstructing 3D scenes from multiple unposed RGB views. MV-DUST3R+ is designed for sparse multi-view RGB input and directly predicts pixel-aligned 3D pointmaps without requiring camera poses or depth maps during inference. The official project describes MV-DUST3R+ as a model that reconstructs large scenes from multiple unposed RGB views, and the CVPR paper introduces cross-reference-view blocks to improve robustness with respect to reference-view selection [2, 1]. This makes MV-DUST3R+ a suitable backbone for sparse-view RGB-only reconstruction.

However, RGB-only reconstruction remains difficult. Appearance alone is often insufficient for metric depth recovery in low-texture areas, occluded regions, repeated structures, and low-overlap views. Monocular depth estimation is also inherently ambiguous because a single image loses metric scale under perspective projection. Therefore, an RGB-only reconstruction backbone may generate visually plausible candidate points that are still geometrically misaligned.

This project addresses the problem by combining RGB-only multi-view reconstruction with estimated depth. The dataset may contain RGB-D frames, but measured depth is not used as an inference-time input. Instead, measured depth is used to fine-tune a monocular depth estimator. At deployment time, the system receives only sparse RGB views. The fine-tuned depth estimator predicts pseudo-depth maps from these RGB images, and the estimated depths are then used to correct the candidate point cloud produced by MV-DUST3R+.

The main motivation is practical deployment. The project aims to use RGB-D training data while preserving an RGB-only inference contract. In other words, depth maps in the dataset teach geometric priors to the model, while the final system does not require the user to provide measured depth.

The main contributions of this report are:

1. It defines a sparse-view RGB-only reconstruction setting in which measured depth is used for training and evaluation, but not as an inference-time input.

2. It proposes a pipeline that combines MV-DUST3R+ with a fine-tuned monocular depth estimator.
3. It describes an estimated-depth correction module that refines candidate geometry produced by the multi-view backbone.
4. It reports both depth-estimation quality and 3D point-cloud reconstruction metrics using the project experiment history.

2. Background and Related Work

This section reviews three related directions: sparse-view RGB-only reconstruction, monocular depth estimation, and depth-assisted 3D reconstruction. These areas motivate the design of the proposed pipeline.

2.1 Sparse-View RGB-Only 3D Reconstruction

Traditional multi-view reconstruction is usually implemented as a multi-stage pipeline consisting of feature detection, feature matching, camera pose estimation, triangulation, bundle adjustment, and dense multi-view stereo. This modular design is interpretable and has been validated in many practical systems. However, each stage depends on the quality of previous stages, so errors can accumulate. These systems also typically require sufficient view overlap, reliable intrinsics, and accurate camera poses.

Recent learning-based methods move toward direct reconstruction from unposed RGB images. DUST3R predicts 3D pointmaps from pairs of RGB images without known camera intrinsics or poses, and MAST3R extends this direction with dense matching grounded in 3D representations [3, 4]. When more than two images are used, pairwise methods must construct many local reconstructions and then align them through global optimization. The MV-DUST3R+ paper notes that pairwise reconstructions may contain matching or geometric errors, and that global optimization cannot always remove these errors [1].

MV-DUST3R handles multiple views in a single feed-forward pass. Instead of exchanging information only between two images, it uses multi-view decoder blocks so that each view can receive information from all other views. MV-DUST3R+ further improves stability by using multiple reference views and cross-reference-view blocks, allowing the model to fuse information across different reference choices [1].

In this project, MV-DUST3R+ is used as the RGB-only multi-view reconstruction backbone. It produces the initial point cloud from a small number of input images. The RGB-only reconstruction is treated as candidate geometry, which can then be checked and corrected using estimated depth.

2.2 Monocular Depth Estimation

Monocular depth estimation predicts a dense depth map from a single RGB image. It is attractive for practical applications because it does not require stereo images or depth sensors. However, the problem is ill-posed. A single image cannot fully determine metric scale because different 3D scenes may project to similar 2D images. Recent work on scale ambiguity in monocular depth also emphasizes that metric depth from one image is under-constrained by perspective projection [8].

Large-scale depth models such as Depth Anything V2 improve monocular depth estimation through strong pretrained encoders and large training corpora. Depth Anything V2 provides robust depth predictions and includes metric-depth variants obtained by fine-tuning pretrained encoders with metric depth labels [5, 6]. This makes it a suitable initialization for the project. The fine-tuning stage uses RGB-D data as supervision so that the depth estimator adapts to indoor scenes and the reconstruction evaluation protocol.

2.3 Depth-Assisted 3D Reconstruction

Depth maps provide strong geometric information for 3D reconstruction. If measured depth and camera poses are available, a point cloud can be constructed by back-projecting valid depth pixels into 3D. However, this assumes that the user has a depth sensor and camera geometry at inference time. This assumption conflicts with the RGB-only deployment goal.

An intermediate solution is to use predicted depth instead of measured depth. Predicted depth is imperfect, but it can still provide useful geometric priors. In this project, predicted depth does not replace the multi-view backbone. Instead, it is used as a correction signal for candidate geometry. This design keeps the multi-view reasoning of MV-DUST3R+ while using learned depth priors to reduce wrong-depth errors.

2.4 Project Positioning

This project does not train a new reconstruction backbone from scratch. Instead, it builds a structured pipeline on top of modern components. MV-DUST3R+ provides candidate geometry from sparse RGB views, while the fine-tuned depth estimator provides a per-image metric-depth prior. The main contribution is the experimental setting, the depth fine-tuning process, and the correction module: using RGB-D data during development while keeping the final inference contract RGB-only.

3. Model Architectures Used in the Project

This section describes the two main model components used in the pipeline: MV-DUST3R+ and Depth Anything V2 Metric Indoor. It is placed before the dataset section so that the reader can understand what each model receives, what it outputs, and how it contributes to the final system.

3.1 MV-DUST3R+ Backbone

MV-DUST3R+ is a feed-forward sparse-view scene reconstruction model. Its input is a set of RGB images. Its output is a set of pixel-aligned 3D pointmaps and associated confidence information. Compared with pairwise DUST3R-style reconstruction, MV-DUST3R+ is designed to reason over multiple input views in one stage. It uses cross-reference-view reasoning to reduce sensitivity to the selected reference view and to improve stability in sparse-view settings.

In this project, MV-DUST3R+ is used only as a frozen or pretrained reconstruction backbone. The project does not claim to modify the MV-DUST3R+ architecture. Instead, MV-DUST3R+ produces the RGB-only candidate point cloud. In the target RGB-only pipeline, the same MV-DUST3R+ reconstruction is also used to recover camera geometry, namely estimated camera intrinsics and poses, from its predicted pointmaps by the pose/intrinsic recovery procedure described by Tang et al. [1]. The proposed method then uses fine-tuned estimated depth and this recovered camera geometry to adjust the candidate point cloud.

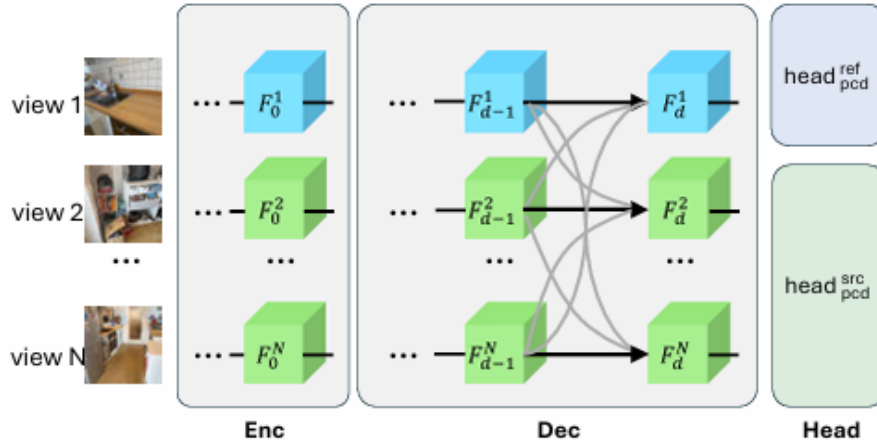


Figure 1: Overview of the MV-DUST3R multi-view reconstruction architecture, adapted from Fig. 3 of Tang et al. [1]. The figure shows shared image encoding, multi-view token fusion in decoder blocks, and pointmap/confidence heads for reference and source views. In this project, this architecture is used as the frozen RGB-only candidate-geometry backbone.

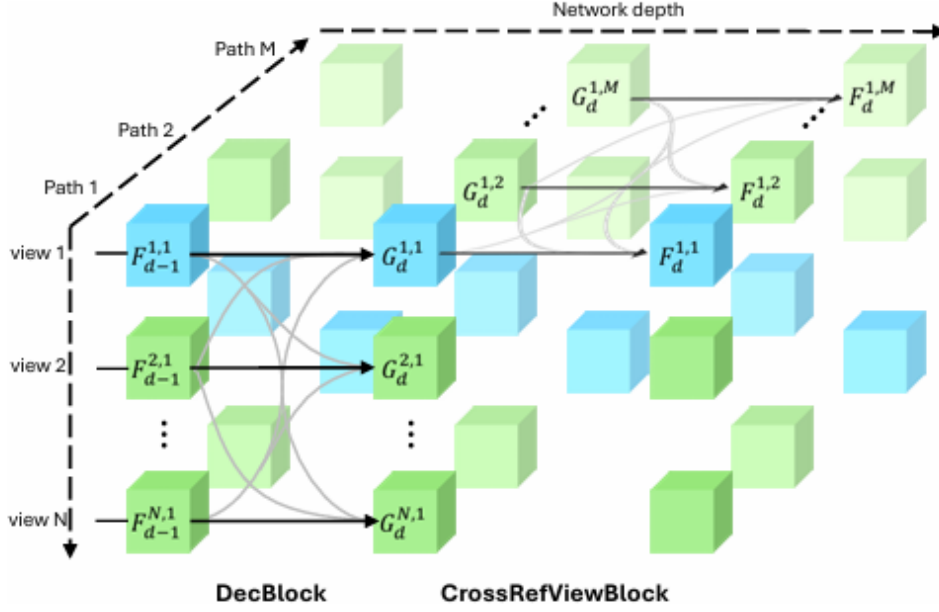


Figure 2: DecBlock and CrossRefViewBlock in MV-DUST3R+, adapted from Fig. 5 of Tang et al. [1]. Multiple reference-view paths are processed in parallel; DecBlocks exchange information among views within each path, while CrossRefViewBlocks fuse the same view across different reference paths.

3.2 Depth Anything V2 Metric Indoor

The depth estimator is initialized from Depth Anything V2 Metric Indoor. This model predicts a dense metric depth map from a single RGB image. The pretrained checkpoint already provides useful indoor depth priors, but the project fine-tunes it on the available RGB-D data so that it better matches the dataset distribution and the reconstruction pipeline.

At inference time, the depth estimator receives each sparse RGB input view independently and predicts an estimated depth map. These estimated depth maps are then used as source-depth anchors for point-cloud correction. Because the depth is predicted from RGB, and the target camera geometry is recovered from the RGB-only MV-DUST3R+ output, the final conceptual pipeline satisfies the RGB-only input contract.

4. Problem Formulation

Let a sparse group contain N RGB views:

$$\mathcal{I} = \{I_1, I_2, \dots, I_N\}, \quad N \in \{3, 4, 5\}.$$

The final inference-time input is only the RGB images. During development, the dataset also provides measured depth maps and camera information that can be used for supervision, diagnostics, and evaluation:

$$\mathcal{D} = \{D_1, D_2, \dots, D_N\}, \quad \mathcal{K} = \{K_1, K_2, \dots, K_N\}, \quad \mathcal{T} = \{T_1, T_2, \dots, T_N\}.$$

The measured depth maps $\mathcal{D}$ are not used as final inference inputs. They are used to fine-tune the depth estimator and to evaluate reconstruction quality. The dataset camera intrinsics and poses, $\mathcal{K}$ and $\mathcal{T}$, are treated as reference camera geometry. In the final RGB-only formulation, the correction module should use camera geometry recovered from the MV-DUS3R+ output:

$$\hat{\mathcal{K}} = \{\hat{K}_1, \hat{K}_2, \dots, \hat{K}_N\}, \quad \hat{\mathcal{T}} = \{\hat{T}_1, \hat{T}_2, \dots, \hat{T}_N\}.$$

The hat symbol means “estimated”, so $\hat{K}_i$ and $\hat{T}_i$ are not ground-truth calibration values provided to the deployed system; they are recovered from RGB-only multi-view reconstruction.

Here, K_i is the reference intrinsic calibration matrix of camera view i , while $\hat{K}_i$ is the estimated intrinsic matrix used by the target RGB-only pipeline:

$$K_i = \begin{bmatrix} f_{x,i} & 0 & c_{x,i} \\ 0 & f_{y,i} & c_{y,i} \\ 0 & 0 & 1 \end{bmatrix}.$$

The estimated matrix $\hat{K}_i$ has the same form, but its entries are recovered by the pipeline instead of read from the dataset. The focal lengths $f_{x,i}, f_{y,i}$ convert normalized camera coordinates to pixel units, while $c_{x,i}, c_{y,i}$ are the principal-point coordinates. Therefore, $\hat{K}_i^{-1}[u, v, 1]^T$ converts an image pixel (u, v) into a camera-ray direction in the local camera frame. This ray vector is a direction vector and is not required to have unit length.

T_i is the reference camera pose for view i , represented as a homogeneous camera-to-world transform. The target pipeline uses the estimated version $\hat{T}_i$:

$$T_i = \begin{bmatrix} R_i & t_i \\ 0 & 1 \end{bmatrix}, \quad R_i \in \mathbb{R}^{3 \times 3}, \quad t_i \in \mathbb{R}^3.$$

The estimated pose $\hat{T}_i$ has the same block structure. The rotation R_i gives the camera orientation in the common world frame, and the translation t_i gives the camera center/location in that frame. In short, $\hat{K}_i$ tells how a 2D pixel becomes a 3D viewing ray, while $\hat{T}_i$ tells where that ray lies in the shared 3D coordinate system. Pose and intrinsic recovery alone do not determine depth: they define the ray, while the estimated depth $\hat{D}_i(u, v)$ chooses a concrete point on that ray.

The MV-DUS3R+ backbone predicts candidate points and provides the information used to recover camera geometry:

$$P_{\text{mv}}, \hat{\mathcal{K}}, \hat{\mathcal{T}} = f_{\text{mv}}(\mathcal{I}).$$

The fine-tuned depth estimator predicts depth maps from RGB:

$$\hat{D}_i = f_{\text{depth}}(I_i).$$

The proposed correction module produces the final point cloud:

$$P_{\text{corr}} = C(P_{\text{mv}}, \hat{\mathcal{D}}, \hat{\mathcal{K}}, \hat{\mathcal{T}}).$$

Historical diagnostic runs may use the dataset K_i, T_i to isolate the effect of estimated depth. The target method, however, replaces measured source depth with estimated depth and replaces externally supplied camera geometry with MV-DUST3R+-recovered camera geometry.

For clarity, the main quantities used throughout the report are:

Quantity	Meaning
N	Number of sparse input views in one group. In this project $N \in \{3, 4, 5\}$.
i	View index, from 1 to N .
I_i	RGB image of view i . This is the only image input required by the final deployment setting.
D_i	Measured depth map of view i , available from the RGB-D dataset. It is used for depth fine-tuning and evaluation, not as final inference input.
$\hat{D}_i$	Estimated depth map predicted from I_i by the fine-tuned monocular depth estimator.
$\mathcal{I}, \mathcal{D}, \hat{\mathcal{D}}, \mathcal{K}, \mathcal{T}$	Collections of RGB images, measured depth maps, estimated depth maps, reference camera intrinsics, and reference camera poses across all sparse views.
$\hat{\mathcal{K}}, \hat{\mathcal{T}}$	Estimated camera intrinsics and poses recovered from the RGB-only MV-DUST3R+ reconstruction. These are the camera quantities used by the target RGB-only correction pipeline.
f_{mv}	Frozen MV-DUST3R+ reconstruction backbone that maps sparse RGB views to candidate 3D pointmaps or point clouds and supports camera-geometry recovery.
f_{depth}	Fine-tuned Depth Anything V2 Metric Indoor depth estimator that maps one RGB image to one metric depth map.
P_{mv}	Candidate point cloud predicted by MV-DUST3R+ before correction.
$C(\cdot)$	Estimated-depth correction module. It compares MV-DUST3R+ candidates with estimated-depth anchors back-projected using $\hat{K}_i$ and $\hat{T}_i$.
P_{corr}	Final corrected point cloud produced by the proposed pipeline.

Table 1: Notation used in the formulation of the proposed RGB-only estimated-depth correction pipeline.

5. Proposed Method

5.1 Pipeline Overview

The proposed pipeline has three stages:

1. **RGB-only candidate reconstruction and camera recovery.** Sparse RGB views are passed into MV-DUST3R+ to obtain candidate 3D points, confidence scores, and recovered camera geometry $(\hat{K}_i, \hat{T}_i)$.
2. **Monocular estimated depth.** Each input RGB image is passed through the fine-tuned Depth Anything V2 Metric Indoor estimator to predict a depth map.

3. **Estimated-depth correction.** Candidate points are compared with points back-projected from estimated depth using the recovered camera geometry. Points with large residuals are moved toward the estimated-depth anchor.

This design is intentionally conservative. It does not replace multi-view reconstruction with monocular depth alone. Instead, depth is used as a correction signal for the multi-view candidate geometry. The role of $\hat{K}_i$ and $\hat{T}_i$ is to turn each estimated depth value into a 3D anchor in the same coordinate frame as the MV-DUS3R+ candidates. These camera quantities define the viewing ray and coordinate frame; the estimated depth chooses the position of the point along that ray.

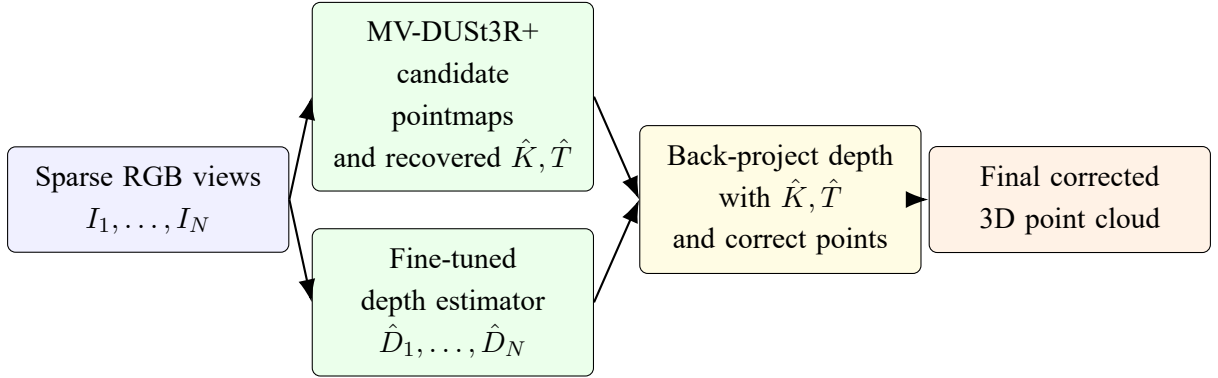


Figure 3: Full target RGB-only pipeline: sparse RGB views are processed by MV-DUS3R+ to obtain candidate pointmaps and recovered camera geometry, while the fine-tuned depth estimator predicts depth maps. Estimated depth is back-projected using $\hat{K}$ and $\hat{T}$, then used to correct candidate geometry and produce the final point cloud.

5.2 Back-Projection from Estimated Depth

For a source pixel (u, v) in view i , the estimated depth is $\hat{z} = \hat{D}_i(u, v)$. First, the pixel is converted into a camera ray using the inverse estimated intrinsic matrix:

$$d_i(u, v) = \hat{K}_i^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}.$$

Multiplying this ray by the estimated depth gives the 3D point in the local camera coordinate system:

$$\hat{X}_{\text{src}}^{\text{cam}}(u, v) = \hat{z} d_i(u, v). \quad (1)$$

This camera-coordinate point is then converted into homogeneous form and mapped into the common world coordinate system by the estimated camera pose $\hat{T}_i$:

$$\begin{bmatrix} \hat{X}_{\text{src}}^{\text{world}}(u, v) \\ 1 \end{bmatrix} = \hat{T}_i \begin{bmatrix} \hat{X}_{\text{src}}^{\text{cam}}(u, v) \\ 1 \end{bmatrix}. \quad (2)$$

Thus, $\hat{K}_i$ controls the direction of the source ray in the camera frame, $\hat{D}_i$ controls how far along the ray the point lies, and $\hat{T}_i$ places that point into the same 3D frame as the MV-DUST3R+ candidate geometry.

In this notation, (u, v) is a pixel coordinate, $\hat{z}$ is the estimated metric depth at that pixel, $d_i(u, v)$ is the corresponding camera ray direction, $\hat{X}_{\text{src}}^{\text{cam}}$ is the estimated-depth point in the local camera coordinate system, and $\hat{X}_{\text{src}}^{\text{world}}$ is the same point after transformation into the common world coordinate system. Without $\hat{D}_i(u, v)$, the recovered pose and intrinsic parameters define only a ray, not a unique 3D point.

5.3 Residual-Gated Correction

Let $X_{\text{pred}}(u, v)$ be the MV-DUST3R+ candidate point associated with the same source pixel. The residual between candidate geometry and estimated-depth geometry is:

$$r(u, v) = \left\| X_{\text{pred}}(u, v) - \hat{X}_{\text{src}}^{\text{world}}(u, v) \right\|_2. \quad (3)$$

The corrected point is:

$$X_{\text{corr}}(u, v) = \begin{cases} (1 - \alpha)X_{\text{pred}}(u, v) + \alpha\hat{X}_{\text{src}}^{\text{world}}(u, v), & r(u, v) \geq \tau, \\ X_{\text{pred}}(u, v), & r(u, v) < \tau. \end{cases} \quad (4)$$

Here, τ is the correction threshold and α is the correction weight. The intended final setting uses estimated depth from the fine-tuned model, not measured RGB-D source depth.

The residual gate has the following interpretation. $X_{\text{pred}}(u, v)$ is the MV-DUST3R+ candidate point associated with pixel (u, v) . The term $r(u, v)$ is the Euclidean distance between this candidate point and the estimated-depth anchor in the world frame. The norm $\| \cdot \|_2$ is the standard 3D Euclidean norm. If $r(u, v) < \tau$, the candidate is already close enough to the estimated-depth geometry and is kept unchanged. If $r(u, v) \geq \tau$, the point is corrected by interpolation. $\alpha = 1$ moves the point fully to the estimated-depth anchor, $\alpha = 0$ keeps the MV-DUST3R+ prediction, and values between 0 and 1 blend the two.

6. Dataset and Experimental Setting

The experiments use a Kaggle-hosted ScanNet-style indoor RGB-D dataset. In the Kaggle environment, the relevant data are discovered under the mounted `posed_images` directory. The dataset is organized by scene. Each scene contains multiple posed RGB-D frames. A frame provides an RGB image, a measured depth map, reference camera intrinsics, and a reference camera pose. The RGB image is the visual input. The measured depth map is the sensor-style depth observation associated with the frame. The reference camera intrinsics K_i describe the projection geometry of view i , and the reference camera pose T_i places that view in a common 3D coordinate system.

The target project formulation does not require these reference camera values as user-

provided inputs at deployment time. Instead, MV-DUST3R+ reconstructs candidate pointmaps from RGB and supports recovery of estimated intrinsics $\hat{K}_i$ and estimated poses $\hat{T}_i$. The reference K_i, T_i remain useful for controlled evaluation, sanity checks, and diagnostic experiments that isolate the effect of depth estimation from the effect of camera-geometry recovery.

The project uses the same data source in two different ways. First, for depth-estimator fine-tuning, the data are treated as a large frame-level RGB-D pool: RGB is the input and measured depth is the supervision target. Second, for 3D reconstruction evaluation, frames are grouped into sparse-view scene-level samples: a group contains only 3, 4, or 5 RGB views from the same scene, and the reconstruction system must produce a 3D point cloud from those sparse views.

6.1 Full Frame Pool for Depth Fine-Tuning

Run 37 discovers the full Kaggle RGB-D frame pool used for fine-tuning the monocular depth estimator. This is larger than the early 30-scene reconstruction subset. The discovered frame-level split is:

Split	Scenes	RGB-D frames	Role in the project
Train	1210	198862	Fine-tune the depth estimator.
Validation	151	24531	Select and report the controlled-best depth checkpoint.
Test	152	25009	Held-out depth quality reporting.
All	1513	248402	Full frame pool discovered from Kaggle.

Table 2: Run 37 frame-level dataset discovery for depth-estimator fine-tuning. The train split is used for optimization; validation and test frames are used to measure depth prediction quality.

This frame-level experiment does not directly evaluate 3D reconstruction. It answers a narrower question: whether the predicted depth maps become accurate enough to serve as geometric anchors for the later estimated-depth correction pipeline.

6.2 Controlled Reconstruction Subset

The completed reconstruction comparisons in this report use a controlled 30-scene subset. The subset was used to validate the reconstruction pipeline before scaling to larger Kaggle coverage. In Run 30, the discovered reconstruction subset contains 30 scenes: 18 train scenes, 6 validation scenes, and 6 test scenes. The validation and test scenes are the evaluated held-out scenes. With `RUN30_MAX_EVAL_GROUPS=0`, the group cap is interpreted as unlimited, so all discovered held-out sparse-view groups are evaluated. The Run 30 rerun records 72 evaluation groups over 12 held-out scenes.

Experiment	Coverage	Use in the report
Run 30	30 scenes; 72 held-out sparse-view groups over 12 validation/test scenes	MV-DUST3R+ candidates corrected with measured RGB-D source depth. This is the validated RGB-D reference.
Run 32	Same controlled reconstruction protocol as Run 30	Direct RGB-D backprojection baseline without MV-DUST3R+. It is a diagnostic baseline, not the proposed method.
Run 33	Reuses Run 30 candidate rows; 36 groups and 72 metric rows	MV-DUST3R+ RGB-only baseline extracted from existing Run 30 candidate rows; no source-depth correction.
Run 37	1513 scenes and 248402 RGB-D frames; frame-level, not group-level	Fine-tunes the depth estimator on the full discovered RGB-D frame pool.

Table 3: Dataset usage by experiment. Scene discovery, group evaluation, and frame-level depth training are separate concepts.

6.3 Sparse-View Group Construction

A sparse-view group is the basic 3D reconstruction sample. All views in one group come from the same scene. The group contains $N \in \{3, 4, 5\}$ selected RGB frames. These RGB frames are passed to MV-DUST3R+ to obtain candidate pointmaps and recovered camera geometry. For correction methods, each source pixel can also be associated with a depth-derived 3D anchor by back-projecting either measured source depth or estimated source depth. In the target RGB-only method, estimated depth is back-projected through $\hat{K}_i$ and $\hat{T}_i$. In oracle or diagnostic RGB-D runs, measured source depth may be back-projected through reference K_i and T_i . For fair point-cloud comparison, each group is evaluated using a fixed point budget; the Run 30 configuration uses at most 3500 candidate points per group.

6.4 Sparse-View Selection Protocol

View selection is part of the controlled evaluation protocol. The project does not solve automatic active view selection; instead, it constructs deterministic sparse-view groups and reuses exactly the same groups for all compared methods. This is important because RGB-only MV-DUST3R+, direct backprojection, measured-depth correction, pretrained-depth correction, and fine-tuned-depth correction must receive the same visual evidence. Therefore, differences in the reported metrics come from the reconstruction method, not from giving one method an easier set of input frames.

For each scene, the group manifest enumerates three view counts, $N = 3$, $N = 4$, and $N = 5$. For each view count, it constructs two view-selection policies:

- **Hybrid policy:** this policy chooses frames that keep moderate local overlap while still covering a wider camera trajectory than a very short temporal window. In the imple-

mentation, it uses fixed frame-index patterns within the first part of the scene sequence, for example a compact but not adjacent set of frames for each N . This policy is intended to avoid both extremes: nearly identical neighboring views and very distant views with too little overlap.

- **Diversity-aware policy:** this policy selects spatially diverse camera positions. It starts from an initial frame and then repeatedly adds the frame whose camera center is farthest from the already selected set. This creates wider-baseline groups and stresses sparse-view reconstruction under lower overlap.

The selected image names are saved in the group manifest as the fixed input definition for that group. If a selection policy accidentally produces duplicate frames, the implementation replaces missing unique views with additional frames from the same scene. This guarantees that a requested 3-view, 4-view, or 5-view group actually contains the requested number of unique RGB inputs.

The evaluation then treats `scene_id + num_views + view_policy` as a stable group key. For example, a group key such as `scene0012_00_4_diversity_aware` means that the scene is `scene0012_00`, the input contains four RGB frames, and the frames were selected by the diversity-aware policy. When the group cap is unlimited, all constructed validation/test group keys are evaluated. When a positive cap is used for a pilot run, groups are sampled in a balanced way across scenes, view counts, and policies.

This design makes the ablation study cleaner. View selection is controlled and shared, while the method under test changes. The report therefore evaluates reconstruction robustness under fixed sparse-view inputs; it does not claim to learn or optimize the camera trajectory itself.

6.5 Use of Measured Depth

Measured depth has three different roles, and the report keeps them separate:

- **Depth supervision:** Run 37 uses measured depth as the training target for the monocular depth estimator.
- **RGB-D reference setting:** Run 30 uses measured source depth directly at inference time. This is a valid RGB-D setting, but it is not RGB-only.
- **Evaluation proxy:** the reconstruction evaluator builds a proxy reference point cloud from the available depth maps and camera geometry. This is useful for controlled comparison, but it is not an independent laser scan or official mesh benchmark.

The target final setting is different from the Run 30 RGB-D reference: it uses RGB images as input, lets MV-DUST3R+ provide both candidate geometry and recovered camera geometry ($\hat{K}_i, \hat{T}_i$), predicts depth with the fine-tuned depth estimator, back-projects the estimated depth using $\hat{K}_i$ and $\hat{T}_i$, and then applies the same style of correction to the MV-DUST3R+ candidate geometry.

6.6 Coverage Terminology and Benchmark Boundary

The report distinguishes three terms:

- **All-scene discovery:** the script scans the mounted Kaggle dataset and discovers all scenes available to that run.
- **All-group metric evaluation:** after scenes are discovered, all constructed sparse-view groups are evaluated when the group cap is set to unlimited.
- **Official benchmark:** an official ScanNet or ScanNet++ benchmark would require the official split, official target representation, official evaluation protocol, and independent benchmark submission or reproduction. This project does not claim such a result.

Therefore, the reported numbers are controlled project metrics on a ScanNet-style Kaggle RGB-D dataset. They are appropriate for comparing the project ablations under the same evaluator, but they should not be presented as official full ScanNet or ScanNet++ benchmark scores.

7. Evaluation Metrics

7.1 Depth Estimation Metrics

Let D be the measured depth map and $\hat{D}$ be the predicted depth map over valid pixels Ω . The depth metrics are:

$$\text{AbsRel} = \frac{1}{|\Omega|} \sum_{p \in \Omega} \frac{|D(p) - \hat{D}(p)|}{D(p)}, \quad (5)$$

$$\text{MAE} = \frac{1}{|\Omega|} \sum_{p \in \Omega} |D(p) - \hat{D}(p)|, \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{p \in \Omega} (D(p) - \hat{D}(p))^2}. \quad (7)$$

The δ_1 accuracy is the fraction of pixels satisfying:

$$\max \left(\frac{\hat{D}(p)}{D(p)}, \frac{D(p)}{\hat{D}(p)} \right) < 1.25. \quad (8)$$

Here, p denotes a valid image pixel, Ω is the set of pixels whose measured depth is valid, $D(p)$ is the measured depth at pixel p , and $\hat{D}(p)$ is the predicted depth at the same pixel. AbsRel measures relative depth error, so a 10 cm error is penalized more strongly for a nearby 50 cm point than for a distant 5 m point. MAE measures the average absolute metric error in meters. RMSE also measures metric error in meters, but it squares the error before averaging and is therefore more sensitive to large mistakes. δ_1 measures the percentage of valid pixels

whose predicted depth is within a multiplicative factor of 1.25 from the measured depth. Lower AbsRel, MAE, and RMSE are better, while higher δ_1 is better.

7.2 Point-Cloud Reconstruction Metrics

Let P be the predicted point cloud and G be the reference point cloud. A point $p \in P$ is one predicted 3D point, a point $g \in G$ is one reference 3D point, and $|P|, |G|$ are the numbers of points in the two clouds. The nearest-neighbor operator $\min_{g \in G} \|p - g\|_2$ finds the closest reference point to p . Accuracy is the average nearest-neighbor distance from prediction to reference:

$$\text{Accuracy} = \frac{1}{|P|} \sum_{p \in P} \min_{g \in G} \|p - g\|_2. \quad (9)$$

Completeness is the average nearest-neighbor distance from reference to prediction:

$$\text{Completeness} = \frac{1}{|G|} \sum_{g \in G} \min_{p \in P} \|g - p\|_2. \quad (10)$$

Accuracy answers: “how far are the predicted points from the reference?” Completeness answers: “how well does the prediction cover the reference?” Both are distances, so lower is better.

Precision and recall are computed at a fixed metric threshold τ_m . In this project $\tau_m = 0.05$ m for the main point-cloud evaluation:

$$\text{Precision} = \frac{|\{p \in P : \min_{g \in G} \|p - g\|_2 \leq \tau_m\}|}{|P|}, \quad (11)$$

$$\text{Recall} = \frac{|\{g \in G : \min_{p \in P} \|g - p\|_2 \leq \tau_m\}|}{|G|}. \quad (12)$$

Precision measures what fraction of predicted points are close enough to the reference. Recall measures what fraction of reference points are recovered by the prediction. F-score is:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (13)$$

Chamfer distance is:

$$\text{Chamfer} = \text{Accuracy} + \text{Completeness}. \quad (14)$$

Chamfer is lower when both the prediction-to-reference and reference-to-prediction distances are small. F-score is higher when both precision and recall are high.

For shape-level analysis, the project also reports Normalized Distance (ND) and the DAc@0.2 metric on selected qualitative groups. Before computing these metrics, both point clouds are center-normalized and scale-normalized. This removes global translation and scale differences so that the metric focuses more on shape agreement. ND is the average normalized nearest-neighbor distance, so lower ND is better. DAc@0.2 is the fraction of normalized points whose nearest-neighbor distance is at most 0.2, so higher DAc@0.2 is better. These normalized

metrics should not be interpreted as absolute metric-pose evaluation.

8. Results

8.1 Depth Estimator Fine-Tuning

Depth evaluation checks whether the fine-tuned depth estimator is accurate enough to serve as a correction anchor. The values in Table 4 and Table 5 are taken from Run 37. The checkpoint used for held-out reporting is `controlled_best`. The `full_dataset_deployment` checkpoint is a deployment artifact because it is trained again on all discovered frames.

Stage	Scenes	Training frames	Epochs	Mini-batch steps	Mean train loss	Runtime
Controlled train split	1210	198862	1	99431	0.0850	4.01 h
Full-data deployment	1513	248402	1	124201	0.0718	4.08 h

Table 4: Run 37 depth-estimator fine-tuning workload. The controlled stage uses the training split to select `controlled_best`; the full-data stage uses all discovered scenes to create the deployment checkpoint.

Depth model	AbsRel ↓	MAE (m) ↓	RMSE (m) ↓	δ_1 ↑	Scale-aligned MAE ↓	Split
Pretrained Depth Anything V2 Metric Indoor	0.2730	0.3839	0.4778	0.6441	0.1672	validation
Fine-tuned <code>controlled_best</code>	0.1145	0.1431	0.2393	0.9355	0.0992	validation
Pretrained Depth Anything V2 Metric Indoor	0.2707	0.3850	0.4806	0.6486	0.1676	test
Fine-tuned <code>controlled_best</code>	0.1140	0.1428	0.2370	0.9362	0.0982	test

Table 5: Run 37 held-out depth evaluation. Each split uses 1200 sampled frames; validation contains 149 scenes and test contains 152 scenes.

Fine-tuning substantially improves depth quality. On validation, AbsRel decreases from 0.2730 to 0.1145, MAE decreases from 0.3839 m to 0.1431 m, and δ_1 increases from 0.6441 to 0.9355. On test, AbsRel decreases from 0.2707 to 0.1140, MAE decreases from 0.3850 m to 0.1428 m, and δ_1 increases from 0.6486 to 0.9362. This indicates that the fine-tuned depth estimator is much more reliable than the initial pretrained checkpoint on the controlled ScanNet-style indoor subset.

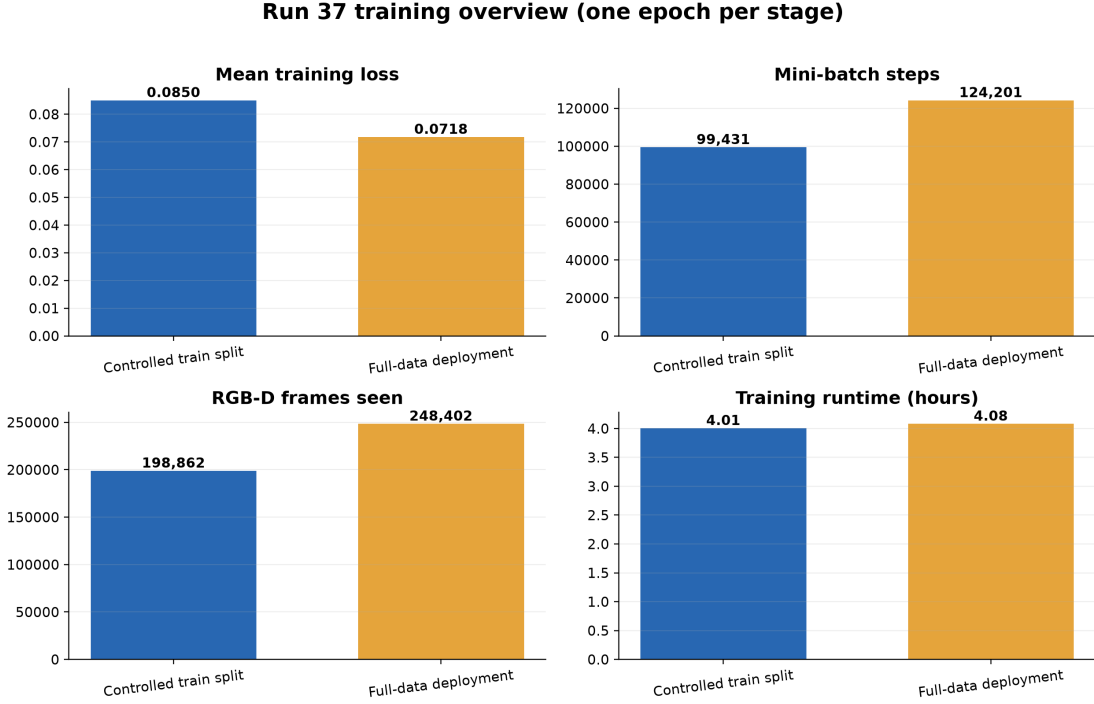
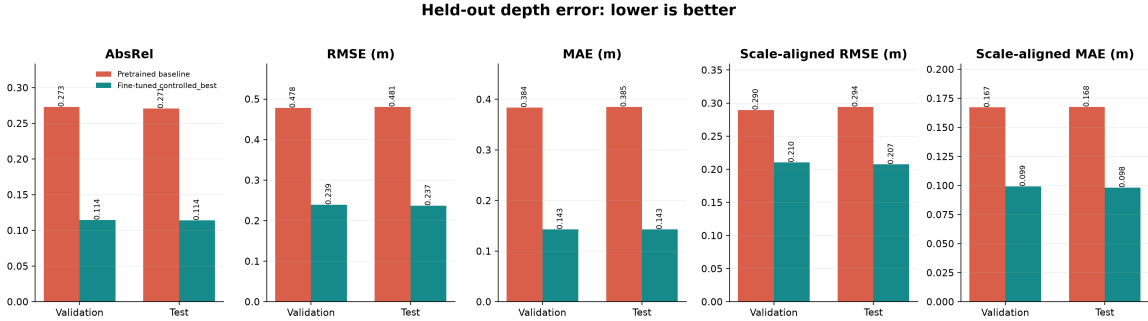
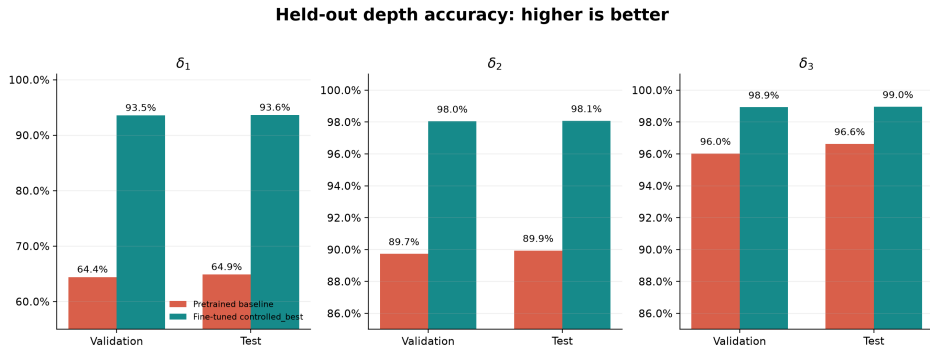


Figure 4: Run 37 fine-tuning workload overview. Because Run 37 logs one epoch per stage, this figure should be interpreted as a workload, mean-loss, and runtime summary rather than as a multi-epoch training-loss curve.



(a) Depth error metrics on validation/test.



(b) Depth accuracy metrics δ_1 , δ_2 , and δ_3 .

Figure 5: Comparison between the pretrained checkpoint and the fine-tuned controlled_best checkpoint.

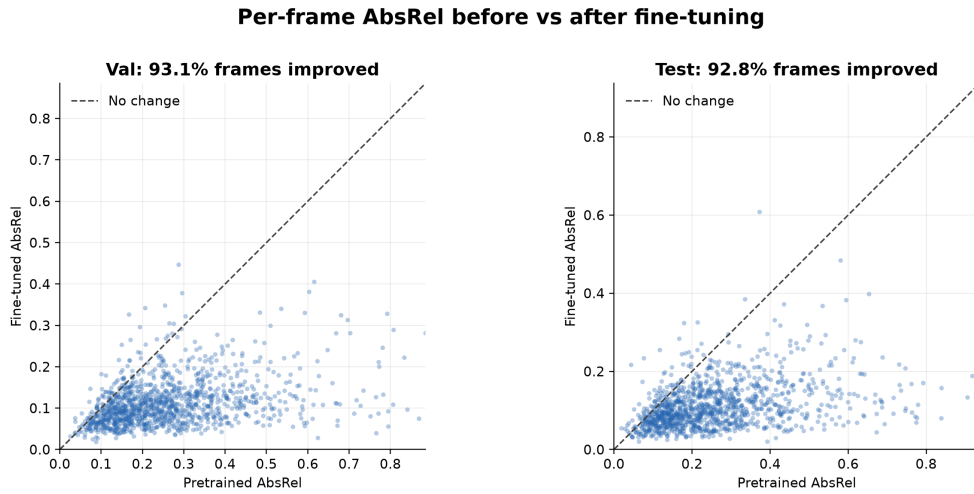
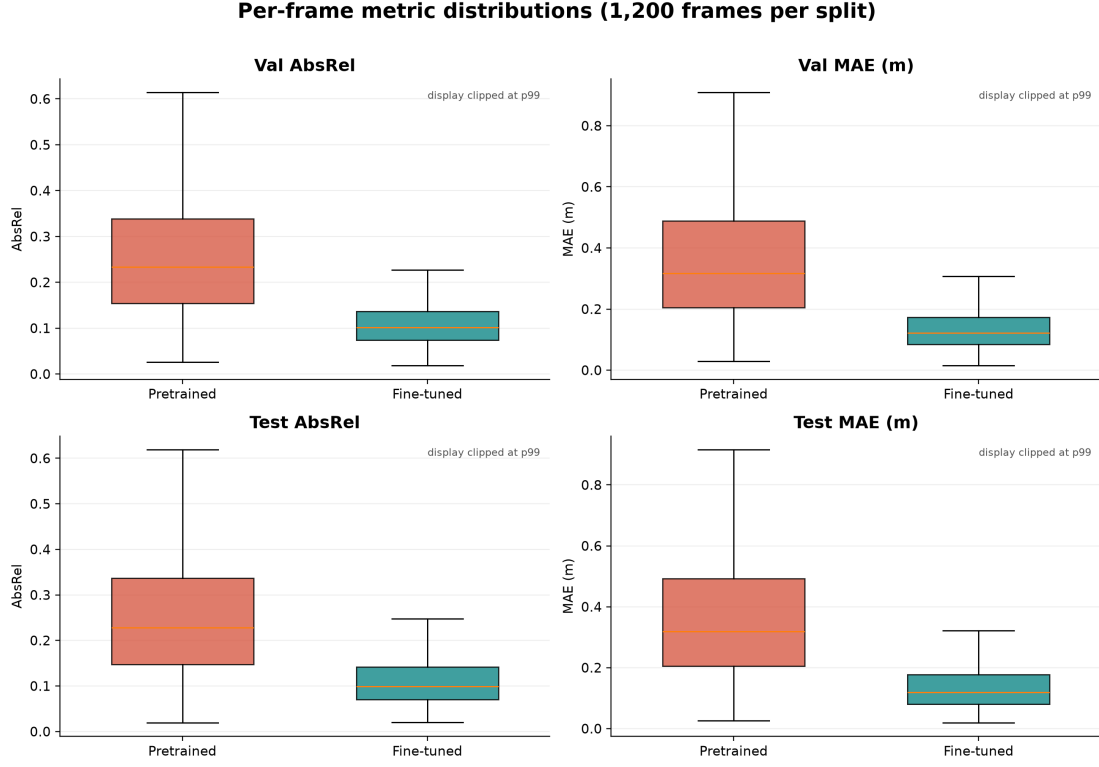


Figure 6: Per-frame analysis shows that the gain is not caused by a few easy frames. The fine-tuned checkpoint reduces AbsRel on about 93.1% of validation frames and 92.8% of test frames.

8.2 3D Reconstruction Results

The main reconstruction experiment evaluates whether fine-tuned estimated-depth correction improves over RGB-only MV-DUST3R+. Table 6 reports the completed Run 41 cached evaluation, which reuses the successful Run 38 and Run 39 cached reconstruction records without rerunning MV-DUST3R+ or the depth estimator. The measured RGB-D source-depth row remains the completed Run 30 oracle reference. These values use the project controlled sparse-

view protocol and should not be interpreted as official full ScanNet or ScanNet++ benchmark results.

Method	Accuracy ↓	Completeness ↓	Precision ↑	Recall ↑	F-score ↑	Chamfer ↓
RGB-only MV-DUST3R+	0.1232	0.1571	0.2915	0.1449	0.1929	0.2803
Direct backprojection from estimated depth	0.1132	0.1037	0.3542	0.2407	0.2862	0.2170
Correction with pretrained estimated depth	0.1648	0.1558	0.1963	0.1356	0.1600	0.3207
Oracle correction with measured RGB-D source depth	0.1272	0.1385	0.3622	0.2283	0.2745	0.2656
Correction with fine-tuned estimated depth (Ours)	0.1422	0.1323	0.2368	0.1445	0.1780	0.2745

Table 6: 3D reconstruction comparison from the completed Run 41 cached evaluation. The measured RGB-D source-depth row is an oracle upper-reference result from Run 30 in the controlled setting, not the RGB-only deployment setting.

The central question is whether the proposed method improves over RGB-only MV-DUST3R+. In the completed Run 41 cached subset, direct backprojection from fine-tuned estimated depth obtains the strongest F-score and Chamfer distance, while fine-tuned estimated-depth correction does not outperform the RGB-only MV-DUST3R+ baseline. The oracle RGB-D row should be interpreted only as a diagnostic upper reference showing what the correction mechanism can do when measured depth is available.

8.3 Ablation Study

The ablation study answers three questions. First, comparing RGB-only MV-DUST3R+ with pretrained-depth correction tests whether off-the-shelf monocular depth helps without task adaptation. Second, comparing pretrained-depth correction with fine-tuned-depth correction tests whether fine-tuning on RGB-D training data improves reconstruction. Third, comparing fine-tuned-depth correction with direct estimated-depth backprojection tests whether MV-DUST3R+ still contributes useful multi-view geometry beyond monocular depth.

The updated ablation set is:

1. **Main:** MV-DUST3R+ candidate geometry + recovered camera geometry + fine-tuned estimated-depth correction.
2. **Ablation 1:** direct 3D backprojection from fine-tuned estimated depth, without MV-DUST3R+ candidate geometry.
3. **Ablation 2:** MV-DUST3R+ candidate geometry + pretrained estimated-depth correction, without depth-estimator fine-tuning.
4. **Ablation 3:** RGB-only MV-DUST3R+ candidate point cloud, without any depth correction.

This analysis is important because the proposed pipeline is not simply a depth-estimation project. It is a reconstruction pipeline that uses estimated depth to correct candidate multi-view geometry.

8.4 Qualitative Analysis

Qualitative results should show reconstructed point clouds side by side. For each selected scene group, the report should include the input RGB views, the RGB-only MV-DUST3R+ point cloud, the direct estimated-depth point cloud, and the final corrected point cloud using fine-tuned estimated depth.

Successful cases are expected to show fewer floating points, better-aligned surfaces, and more stable geometry in ambiguous regions. Failure cases should also be reported, especially scenes with severe occlusion, repeated furniture, reflective surfaces, or inaccurate estimated depth.

9. Discussion

The proposed project is motivated by deployability. A system that requires measured depth at inference time can be accurate, but it is limited to users with specialized sensors. A system that only requires RGB images is easier to deploy because most cameras produce RGB images by default. Therefore, this project uses RGB-D data for training and evaluation while keeping the final inference input RGB-only.

This design also changes the interpretation of the dataset. The depth maps are not ignored. They are used as supervision for the depth estimator and as reference information for reconstruction evaluation. The final system then applies the learned depth knowledge to RGB-only inputs.

The expected benefit over RGB-only MV-DUST3R+ is geometric stability. MV-DUST3R+ provides multi-view candidate geometry but may still predict wrong depth in sparse-view settings. The fine-tuned depth estimator provides a per-image geometric prior. When estimated depth is reliable, the correction module can move candidate points toward more plausible metric positions. When estimated depth is unreliable, the correction should be weakened or rejected. This motivates future work on confidence-aware depth correction.

The second important factor is camera-geometry recovery. In the target RGB-only pipeline, the correction anchor depends on $\hat{K}_i$, $\hat{T}_i$, and $\hat{D}_i$. If the recovered camera pose or intrinsic matrix is inaccurate, even a good depth estimate can be back-projected to the wrong 3D position. Therefore, diagnostic runs using reference K_i , T_i are useful for isolating depth quality, but they should not be confused with the final RGB-only camera setting.

The main limitation is that monocular depth is still imperfect. Fine-tuning reduces domain mismatch, but it does not remove all ambiguity. The depth model may still fail on unusual layouts, transparent objects, reflective surfaces, and scenes whose scale distribution differs from the training data. Camera-geometry recovery can also introduce errors in low-overlap or ambiguous views. Therefore, final claims should remain limited to the controlled sparse-view RGB-only setting and should not be presented as an official ScanNet or ScanNet++ benchmark unless the official protocol is implemented.

10. Conclusion

This project studies sparse-view RGB-only 3D scene reconstruction from a small number of ordinary 2D images. The motivation is practical: most everyday images do not include measured depth, and requiring a depth sensor would reduce deployability. The project therefore uses measured RGB-D data for supervision and evaluation, but not as an inference-time input.

The proposed pipeline combines MV-DUST3R+ with a fine-tuned monocular depth estimator. MV-DUST3R+ produces candidate multi-view geometry and recovered camera geometry from sparse RGB views, while the depth estimator predicts pseudo-depth maps from the same RGB inputs. The estimated-depth correction module back-projects those depth maps using the recovered camera intrinsics and poses, then refines candidate geometry to reduce wrong-depth errors and improve point-cloud quality.

The report defines an evaluation framework for this setting. Depth quality is measured using monocular depth metrics, while 3D reconstruction quality is measured using point-cloud metrics such as accuracy, completeness, precision, recall, F-score, and Chamfer distance. The main comparisons are RGB-only MV-DUST3R+, direct estimated-depth backprojection, pretrained-depth correction, and fine-tuned-depth correction. The final fine-tuned method should only be selected if it improves reconstruction metrics while preserving RGB-only inference.

Future work should evaluate more sparse-view groups under the updated camera-geometry protocol, add confidence-aware correction, and test the method on more diverse real RGB image sets. The long-term goal is a deployable reconstruction system that can produce useful 3D scene geometry from only a few ordinary 2D images.

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